LAB 1

Daily Sales Analysis with NumPy

Python / NumPy • Google Colab or Jupyter

Duration	30–60 minutes
Objectives	Create NumPy arrays; inspect data; calculate statistics; filter values; extract simple business insights.
Tools	Google Colab or Jupyter Notebook; Python 3; NumPy
Submission	One notebook containing code, outputs, and short written answers.

Scenario

You are helping a small shop analyze its sales for one working week. The manager wants simple answers: How much did we sell? What was the average? Which day performed best? Which days were above a target?

Dataset

Use the following data throughout the lab:

```
import numpy as np

days = np.array(["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"])
sales = np.array([120, 150, 95, 180, 160, 210, 175])
```

Exercise 1 — Create and inspect the sales data

Goal: become comfortable with NumPy arrays, indexing, shape, and size.

1. Import NumPy using the alias np.
2. Create the days and sales arrays using the dataset above.
3. Print the sales array.
4. Print the first sales value and the last sales value.
5. Print sales.shape and sales.size.
6. Print the sales values from Tuesday to Friday using slicing.

Expected skills: np.array(), indexing with [ ], slicing with [start:end], .shape, and .size.

Checkpoint: Before moving on, verify that your first value is 120 and your last value is 175.

Exercise 2 — Calculate weekly performance

Goal: use NumPy to calculate basic statistics without writing loops.

7. Calculate the total weekly sales using .sum().
8. Calculate the average daily sales using .mean().
9. Find the minimum and maximum sales using .min() and .max().
10. Find the index of the best sales day using .argmax().
11. Use that index with the days array to display the name of the best sales day.

Then answer in one or two sentences:

- What was the total weekly sales?
- What was the average daily sales?
- Which day had the highest sales?

Exercise 3 — Filter the data and extract insights

Goal: use comparisons and Boolean masks to answer simple business questions.

12. Create a condition for sales greater than 150.
13. Display only the sales values greater than 150.
14. Display the names of the days where sales were greater than 150.
15. Create a second condition for sales greater than or equal to the weekly average.
16. Display the days whose sales were greater than or equal to the average.
17. Count how many days had sales greater than 150.

Business questions:

- How many days exceeded 150 in sales?
- Which days performed at or above the weekly average?
- Write one short conclusion for the shop manager.

Final challenge — Clean notebook

Organize your notebook so that another student can understand it easily. Use Markdown headings for the three exercises, meaningful variable names, and short comments only where they help explain the code.

Your final notebook should contain:

- A title and short description of the analysis.
- The dataset and NumPy import.
- Solutions for Exercises 1–3.
- Visible outputs for the important calculations.
- Short answers to the business questions.

Beginner tips

- Remember: NumPy indexing starts at 0.

- sales[0] is the first value; sales[-1] is the last value.
- sales[1:4] includes indices 1, 2, and 3.
- A condition such as sales > 150 creates True/False values.
- sales[sales > 150] keeps only the values that satisfy the condition.